Solution Architecture

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Team ID	6ab63662000bf9bb96385cb4
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	3 Marks

Solution Architecture

This document describes the high-level architecture of the Auto Ticket Classification system, including all major components and their interactions within the ServiceNow platform.

Architecture Layers

Layer 1 - Presentation Layer (User Interface):
- Service Portal: End-user incident submission form with guided inputs
- Agent Workspace: Enhanced incident view with classification confidence display
- PA Dashboard: Real-time classification metrics and accuracy trends

Layer 2 - Automation Layer (Flow Designer):
- Main Flow: 'Auto Classify Incident' (trigger: Incident After Insert)
- Subflow: 'Invoke Predictive Intelligence' (calls PI classification API)
- Subflow: 'Route to Assignment Group' (maps category to group)
- Subflow: 'Confidence Threshold Check' (>= 80% auto, < 80% manual)
- Subflow: 'Escalation Handler' (P1/P2 immediate notification)

Layer 3 - Intelligence Layer (Predictive Intelligence):
- ML Classification Model trained on historical incident data
- Input: Short Description + Description text fields
- Output: Category, Subcategory, Assignment Group with confidence %
- Retraining Pipeline: Weekly scheduled job

Layer 4 - Data Layer:
- Incident Table [incident]: Core incident records
- sys_prediction: PI prediction results and confidence scores
- sys_prediction_model: Trained model metadata
- Assignment Group Table [sys_user_group]: Resolver group mappings
- CMDB [cmdb_ci]: Configuration item relationships

Component Interaction Flow

1. End User submits incident via Service Portal / Email
2. Incident record created in Incident Table
3. Business Rule triggers Flow Designer Main Flow
4. Flow calls Predictive Intelligence classification API

5. PI model returns predicted Category, Subcategory, Assignment Group + confidence score
6. Flow checks confidence threshold:
   a. >= 80%: Auto-populates fields, sets state to 'In Progress', notifies assigned group
   b. < 80%: Adds work note, routes to L1 manual triage queue
7. If Priority = P1/P2: Escalation subflow triggers (page on-call, email manager)
8. Incident updated in database with classification audit trail
9. Performance Analytics ingests data for dashboard reporting

Security & Access Control

- All processing occurs within the ServiceNow instance (no external data transfer)
- Role-based access: itil role for agents, admin for PI model management
- Audit logs capture all auto-classification actions and manual overrides
- Update Sets used for change management across instances (Dev -> Test -> Prod)